

MCU (STM32WB55), SWD Debug & RF (BLE) — BLEC-SCH-0002

Essential performance (controller role):

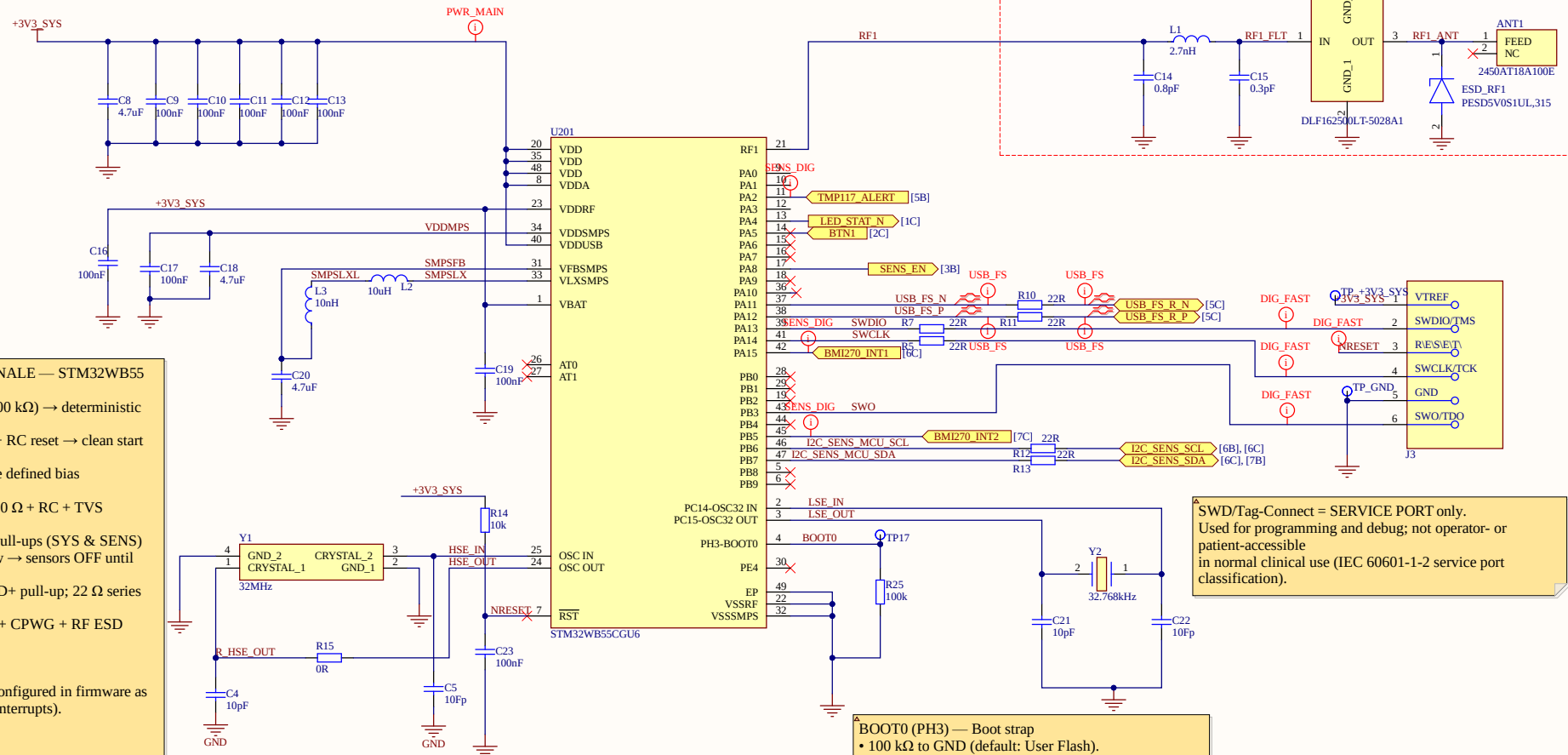
- Board provides BLE command/control only; no direct patient-applied parts.
- All MCU I/Os are SELV and default to safe states on reset (pull-ups/-downs and series resistors as shown).
- BLE link integrity and stimulation safety are handled at system/implant level.

IEC 60601 Compliance:

USB-C = AC/DC SELV Input Port.
PPTC + TVS + ESD + CMC provide protection for IEC 61000-4-2/4-4/4-5.
All voltages SELV; battery safety provided by charger + pack.

RF port (intentional radiator):

2.4 GHz BLE antenna with π -match (C-L-C) and RF ESD footprint.
CPWG, via fence and π -match used to meet radiated emission/immunity requirements (IEC 60601-1-2).



MCU DESIGN RATIONALE — STM32WB55

- BOOT0 pulled low (100 k Ω) $\rightarrow$ deterministic User Flash boot
- NRST has 10 k Ω PU + RC reset $\rightarrow$ clean start under EMC
- All interrupt lines have defined bias (TMP117/BMI)
- BTN1 has pull-up + 100 Ω + RC + TVS (false-trigger resistant)
- I²C buses use 4.7 k Ω pull-ups (SYS & SENS)
- SENS_EN defaults low $\rightarrow$ sensors OFF until MCU enables
- USB FS uses internal D+ pull-up; 22 Ω series resistors
- RF path uses π -match + CPWG + RF ESD footprint

Unused GPIO Policy:
All unused MCU pins configured in firmware as ANALOG (no pull, no interrupts).

BOOT0 (PH3) — Boot strap

- 100 k Ω to GND (default: User Flash).
- TP to +3V3 to force System Bootloader.
- Do not use as GPIO.

SWD/Tag-Connect = SERVICE PORT only.

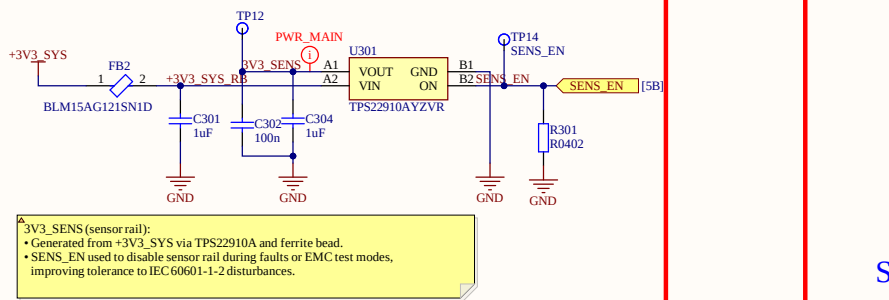
Used for programming and debug; not operator- or patient-accessible in normal clinical use (IEC 60601-1-2 service port classification).

Title BLE-Control Wearable — Schematic Set (AD25)

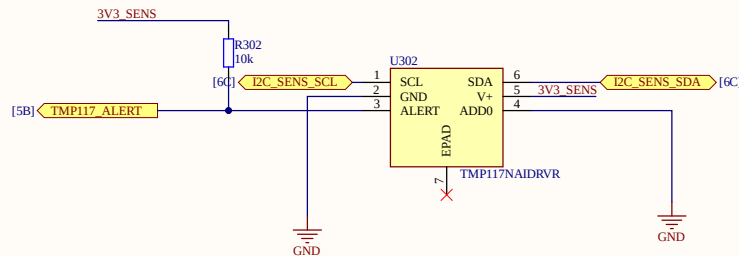
Size	Number	Revision
A3	BLEC-SCH-0002	A0 (EVT)
Date:	5/26/2026	Sheet of 2 of 3
File:	C:\Users\MCU_RF\SchDoc	Drawn By: Caoilte Donohoe

Sensors/IO: TPS22910A 3V3_SENS, Button, LEDs, I²C — BLEC-SCH-0003

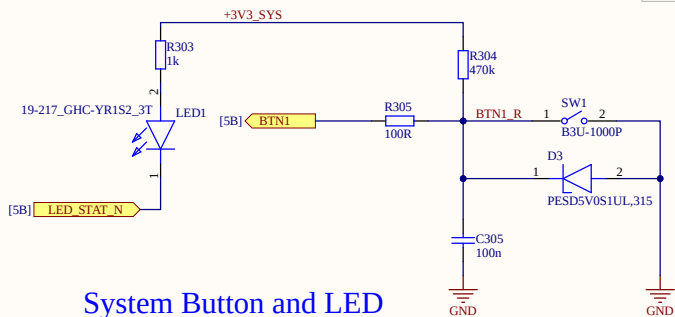
Sensor Power Gate



Skin Temperature Sensor

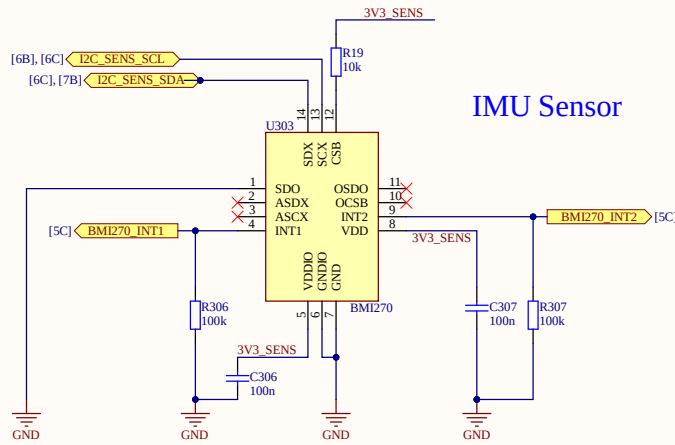


System Button and LED



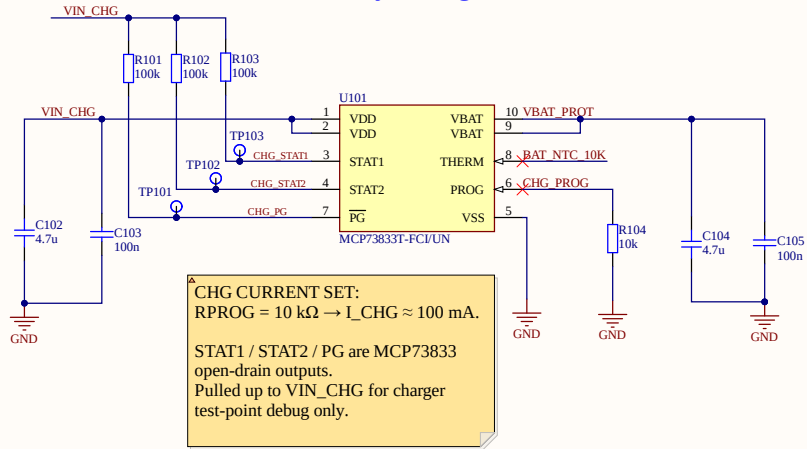
A Operator I/O (button + LED + skin sensor):
• Button: TVS at pad + 100R series + RC network for ESD and burst immunity.
• Sensors: on-board only, short I2C, powered via 3V3_SENS gate.
• No direct patient-applied parts on this PCB; temperature is read through enclosure/assembly.
• BTN1 default state:
• Pull network ensures defined logic level on reset and under noise.

IMU Sensor

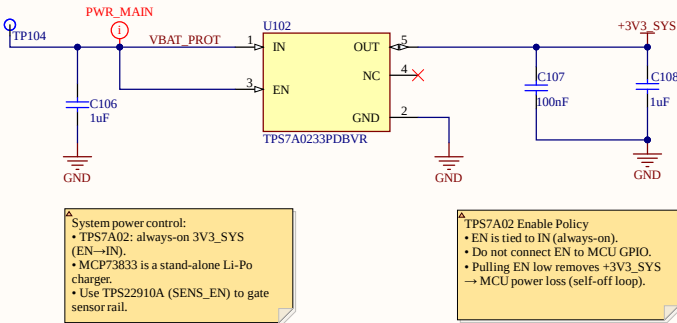


Power, Battery, USB-C & MCP73833T (Main 3V3) — BLEC-SCH-0001

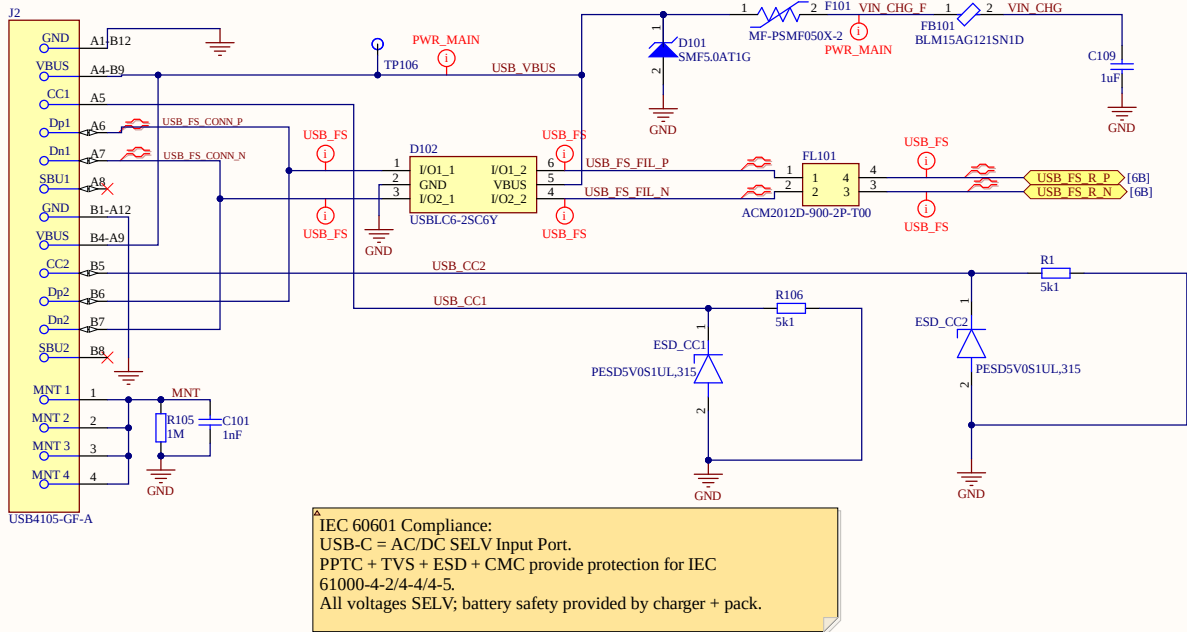
Battery Charge Cct



REG LINEAR



USB Filtering and ESD



Battery Connect

